

Astroinformatics 1: Graded Practice 4

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I. GITHUB REPOSITORY

The first objective of this practice was to create a GitHub repository to store all the material utilized during the graded practice of the Astroinformatics course. This was done via the GitHub web page for ease of use. However it should be noted that due to this method, a lot of the light curves CSV and LC files had to be uploaded by batches of 10 files, as the GitHub webpage has a strict limit on the size of the commit that can be pushed into the repository.

Following the creation of the repository and the uploading of all the course material, basic documentation for the workflow of the TESS light curves data was written in the README file, explaining the general structure on how the raw data obtained during the graded practice 1, was later transformed into readable light curves on the graded practice 3.

II. TEST CASES FOR TESS LIGHT CURVES PLOTTING

As explained on the README file, the data processing workflow of the TESS light curve data was divided into three main stages, which roughly correspond to the graded practices 1, 2 and 3, the acquisition, format transformation, and statistical analysis. Some Test Cases that could be applied to this data pipeline are the following:

1. Format Transformation Integrity:

- Input: A standard TESS CSV file.
- Expected Output of the TASK.SH shellscript: A .lc file with space delimiters containing exactly 6 columns with the correct headers (TIME, SAP FLUX, etc.)

2. Coarse Frequency Grid Spacing (Lomb-Scargle Misuse)

- Input: The user manually specifies a frequency grid that is too coarse (e.g., only evaluating 100 points over a large range) instead

of using the `autopower()` heuristic or an appropriately fine grid.

- Expected Output: If the grid points are too far apart, significant periodic peaks can fall into the gaps between points and be missed entirely. Therefore, the resulting periodogram will erroneously show very low power across all frequencies, leading the user to the false conclusion that no periodic signal exists in the data, even if a strong one is present.

3. Robust Outlier Identification:

- Input: Data with known extreme spikes (for example, Figure 2 in Practice 3).
- Expected Output: The script should flag points exceeding $3 \times (1.4826 \times \text{MAD})$ as outliers and render them with distinct red "star" markers in the final plot.

4. Invalid or Default Directory Path

- Input: Running the formatting shell script (`task.sh`) from Practice 2 without updating the SOURCE DIR variable (leaving it as the placeholder "[Insert here the path]").
- Expected Output: The script uses a for loop to look for *.csv files specifically within that defined directory path. So, The script will fail to find any files. The terminal will not display any "Created .lc" messages, and no new files will be generated in the working directory because the loop for file in "SOURCE DIR"/*.csv; do will have no matches to iterate over.

5. Periodicity Accuracy:

- Input: A light curve with a clear sinusoidal or cosine signal (e.g., Figures 1 and 3 in Practice 3).
- Expected Output: The Lomb-Scargle periodogram should show a high-power peak at the correct period, and the folded light curve should show a coherent, non-random shape.

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